Chemical reactions

30 questions: 15 multiple choice, 15 structured.

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1 Core Physical and chemical changes [1]

Which change is a chemical change?

A rusting of an iron nail

B melting of candle wax

C dissolving of sugar in water

D evaporation of ethanol

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2 Core Rate of reaction [1]

Marble chips react with excess dilute hydrochloric acid to produce carbon dioxide gas. The experiment is repeated at a higher temperature, with all other conditions kept the same. What happens to the rate of the reaction?

A It increases

B It decreases

C It stays the same

D It increases at first, then decreases to zero

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3 Core Rate of reaction [1]

Which statement about a catalyst is correct?

A A catalyst is used up during the reaction it speeds up

B A catalyst increases the rate of a reaction and is chemically unchanged at the end of it

C A catalyst decreases the rate of a reaction

D A catalyst changes the products formed by a reaction

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4 Core Rate of reaction [1]

A student investigates the reaction between calcium carbonate lumps and excess dilute hydrochloric acid, which produces carbon dioxide gas. Which apparatus set-up allows the volume of gas produced to be measured directly at regular time intervals?

A a thermometer placed in the reaction mixture

B a gas syringe connected to the reaction flask

C a balance placed under the open reaction flask

D a pH probe placed in the reaction mixture

5 Supplement Rate of reaction [1]

According to collision theory, how does a catalyst increase the rate of a reaction?

A It increases the number of reactant particles per unit volume

B It increases the average kinetic energy of the reactant particles

C It provides an alternative reaction pathway with a lower activation energy

D It increases the pressure of the reacting system

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6 Core Rate of reaction [6]

A student reacts excess small marble chips (calcium carbonate) with dilute hydrochloric acid in a conical flask plugged with cotton wool, standing on a balance. The flask and contents are weighed every 30 seconds as carbon dioxide gas escapes. The table shows the results.

Time / s	**0**	**30**	**60**	**90**	**120**	**150**	**180**
Total mass lost / g	0.00	0.32	0.56	0.72	0.82	0.88	0.88

(a) State two variables that should be kept constant during this experiment for it to be a fair test. [2]

(b) Calculate the mean rate of reaction, in g/s, over the first 60 seconds. [2]

(c) Describe how the rate of the reaction changes between 0 s and 180 s, and explain what the results show has happened to the reaction by 150 s. [2]

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7 **Supplement** Rate of reaction [6]

Dilute hydrochloric acid reacts with small pieces of magnesium ribbon.

(a) Using collision theory, explain why increasing the concentration of the hydrochloric acid increases the rate of this reaction. [3]

(b) Using collision theory, explain why increasing the temperature of the hydrochloric acid increases the rate of this reaction. [3]

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8 **Supplement** Rate of reaction [5]

A teacher wants to compare two methods of investigating the rate of reaction between marble chips and dilute hydrochloric acid: Method 1 measures the loss in mass of the reaction flask over time using a balance; Method 2 measures the volume of gas collected in a gas syringe over time.

(a) State one piece of data that both methods can be used to calculate. [1]

(b) Evaluate Method 1, compared with Method 2, for this particular reaction. [2]

(c) Suggest one improvement that could be made to Method 1 to reduce error in the results. [2]

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9 **Core** Reversible reactions and equilibrium [1]

Which symbol shows that a reaction is reversible?

A ⇌

B →

C =

D ±

10 Core Reversible reactions and equilibrium [1]

White anhydrous copper(II) sulfate is used to test for the presence of water. What colour does it turn when water is added?

A pink

B yellow

C blue

D it stays white

11 Supplement Reversible reactions and equilibrium [1]

Which equation correctly represents the reaction taking place in the Haber process?

A $N_2(g) + H_2(g) \rightleftharpoons NH_3(g)$

B $2N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

C $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

D $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

12 Supplement Reversible reactions and equilibrium [1]

What are the typical conditions used in the Haber process?

A 450 °C, 200 kPa, vanadium(V) oxide catalyst

B 450 °C, 20 000 kPa, iron catalyst

C 200 °C, 20 000 kPa, platinum catalyst

D 450 °C, 20 000 kPa, nickel catalyst

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13 **Supplement** Reversible reactions and equilibrium [1]

In the Contact process, which is a typical source of the sulfur dioxide used?

A cracking crude oil fractions

B fractional distillation of liquid air

C electrolysis of concentrated sodium chloride solution

D burning sulfur or roasting sulfide ores

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14 **Supplement** Reversible reactions and equilibrium [7]

Nitrogen and hydrogen react in a closed container: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, $\Delta H = -92$ kJ/mol (the forward reaction is exothermic).

(a) State the two conditions that show a reversible reaction in a closed system has reached equilibrium. [2]

(b) Predict and explain the effect of increasing the pressure on the position of this equilibrium. [2]

(c) Predict and explain the effect of increasing the temperature on the position of this equilibrium. [2]

(d) State the effect of adding a catalyst on the position of this equilibrium. [1]

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15 **Supplement** Reversible reactions and equilibrium [6]

The Haber process is used industrially to manufacture ammonia.

(a) Write the balanced symbol equation, including state symbols, for the Haber process. [2]

(b) State the raw material source of the hydrogen and the raw material source of the nitrogen used in the process. [2]

(c) State the typical temperature and pressure used in the Haber process, and name the catalyst used. [2]

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16 **Supplement** Reversible reactions and equilibrium [6]

The Contact process is used industrially to manufacture sulfur trioxide, which is later converted into sulfuric acid.

(a) Write the balanced symbol equation, including state symbols, for the conversion of sulfur dioxide to sulfur trioxide. [2]

(b) State a source of the sulfur dioxide and the source of the oxygen used in the process. [2]

(c) State the typical temperature and pressure used in this stage of the Contact process, and name the catalyst used. [2]

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17 **Supplement** Reversible reactions and equilibrium [6]

The forward reaction in the Haber process is exothermic, and produces fewer gas molecules than it uses. Industrially, a temperature of 450 °C and a pressure of 20 000 kPa are used, rather than a much lower temperature or a much higher pressure.

(a) Explain, in terms of rate of reaction and position of equilibrium, why a compromise temperature of 450 °C is used rather than a much lower temperature. [3]

(b) Explain, in terms of position of equilibrium and cost/safety, why a pressure of 20 000 kPa is used rather than a much higher pressure. [3]

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18 **Supplement** Reversible reactions and equilibrium [6]

Blue anhydrous cobalt(II) chloride paper turns pink when water is added to it, and the pink paper turns back to blue on heating. This can be summarised as a reversible reaction:

hydrated cobalt(II) chloride (pink) ⇌ anhydrous cobalt(II) chloride (blue) + water

(a) State what is meant by describing this reaction as reversible. [2]

(b) Predict and explain what you would observe if water is added to a sample of the blue anhydrous solid. [2]

(c) Predict and explain what you would observe if the pink hydrated solid is heated strongly. [2]

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19 Core Redox [1]

What is the oxidation number of iron in iron(III) oxide, Fe_2O_3?

A +2

B +6

C −3

D +3

20 Core Redox [1]

In the reaction $CuO(s) + H_2(g) \rightarrow Cu(s) + H_2O(l)$, which statement is correct?

A CuO is reduced because it loses oxygen

B CuO is oxidised because it loses oxygen

C H_2 is reduced because it gains oxygen

D H_2 is reduced because it loses oxygen

21 Core Redox [1]

In the reaction $Mg(s) + CuSO_4(aq) \rightarrow MgSO_4(aq) + Cu(s)$, which species is oxidised?

A Cu^{2+}

B SO_4^{2-}

C Cu

D Mg

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22 Supplement Redox [1]

What is the oxidation number of nitrogen in the nitrate ion, NO_3^-?

A +3

B −3

C +5

D +6

23 Supplement Redox [1]

Acidified potassium manganate(VII) solution is added to a colourless aqueous solution of potassium iodide. Which colour change is observed?

A colourless to purple

B purple to (almost) colourless, with a brown colour developing in the solution

C orange to green

D no colour change occurs

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24 Core Redox [4]

Iron(III) oxide reacts with carbon monoxide in a blast furnace: $Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(l) + 3CO_2(g)$

(a) Identify the substance that is reduced in this reaction, giving a reason based on gain or loss of oxygen. [2]

(b) Identify the substance that is oxidised in this reaction, giving a reason based on gain or loss of oxygen. [1]

(c) Explain why this reaction is described as a redox reaction. [1]

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25 Core Redox [4]

Iron forms two common oxides, FeO and Fe_2O_3. Copper forms two common oxides, Cu_2O and CuO. Give the full name of each oxide below, using a Roman numeral to show the oxidation number of the metal.

(a) FeO [1]

(b) Fe_2O_3 [1]

(c) Cu_2O [1]

(d) CuO [1]

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26 **Supplement** Redox [5]

Zinc metal is added to aqueous copper(II) sulfate: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

(a) Define oxidation in terms of electron transfer. [1]

(b) Write an ionic half-equation, including state symbols, for the change happening to the zinc. [2]

(c) State, with a reason based on electron transfer, whether the copper(II) ion is oxidised or reduced. [2]

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27 **Supplement** Redox [6]

Use the rules for oxidation numbers to answer the following.

(a) State the oxidation number of nitrogen in the ammonium ion, NH_4^+. [1]

(b) State the oxidation number of chromium in the dichromate ion, $Cr_2O_7^{2-}$. [2]

(c) State the oxidation number of sulfur in the sulfite ion, SO_3^{2-}. [2]

(d) State the oxidation number of manganese in the manganate(VII) ion, MnO_4^-. [1]

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28 **Supplement** Redox [7]

Acidified aqueous potassium manganate(VII) is purple. It is added to an excess of colourless aqueous potassium iodide, and the manganate(VII) ions are converted to pale pink/colourless manganese(II) ions while the iodide ions are converted to brown aqueous iodine.

(a) State the colour change observed in this reaction. [1]

(b) Define an oxidising agent, and identify the oxidising agent in this reaction. [2]

(c) Define a reducing agent, and identify the reducing agent in this reaction. [2]

(d) State whether the manganate(VII) ion is oxidised or reduced, giving a reason in terms of oxidation number. [2]

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29 **Supplement** Redox [6]

In the blast furnace, iron(III) oxide reacts with carbon monoxide: $Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(l) + 3CO_2(g)$

(a) Define an oxidising agent and a reducing agent. [2]

(b) Identify the oxidising agent and the reducing agent in this reaction, giving a reason for each. [4]

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30 **Supplement** Redox [8]

Aqueous iron(II) sulfate reacts with chlorine gas: $2Fe^{2+}(aq) + Cl_2(g) \rightarrow 2Fe^{3+}(aq) + 2Cl^-(aq)$

(a) State the oxidation number of iron before and after this reaction. [2]

(b) State the oxidation number of chlorine before and after this reaction. [2]

(c) Identify which species is oxidised and which is reduced, using the oxidation number changes from (a) and (b). [2]

(d) Write separate ionic half-equations for the oxidation process and the reduction process taking place. [2]

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